

Module 11: National and International Technology Transfer Policies

National Technology Transfer Regulations (NTTR):

National Technology Transfer Regulations are rules and guidelines made by a country to manage how technology is shared, especially from public research institutions (like universities and government labs) to private companies or even other countries. These regulations make sure that technology transfer helps economic growth, supports innovation, and protects national security.

Key Objectives of NTTR

1. **Promote Commercialization of R&D** – Encourage universities and public research centres to share their developed technologies with industries so they can be used in the market.
2. **Protect National Interests** – Stop the transfer of sensitive or defense-related technologies that could harm national security if leaked.
3. **Enable Collaboration** – Create partnerships between government, universities, and industries to drive innovation.
4. **Revenue Generation** – Allow public research bodies to earn money through patents, licenses, and technology transfer.
5. **Ensure Fair Access** – Help startups and small businesses (MSMEs) get access to technologies at affordable and fair prices.

Key Components of National Technology Transfer Regulations:

Component	Description
Technology Licensing	Lets research institutions give licenses of their patented technologies to private companies.
IPR Ownership Rules	Specifies who owns the intellectual property (IP) developed with public funding.
Disclosure Norms	Researchers must report any inventions made with government funding.
Evaluation Committees	Expert groups check the value and usefulness of technology before transfer.
Foreign Collaboration Rules	Restricts or governs tech transfer to foreign entities.
Royalty & Revenue Sharing	Sets terms for sharing revenue from licensed technologies among stakeholders.

Example: Bangladesh's Technology Transfer Framework

Bangladesh does not have a single unified “National Technology Transfer Act,” but several frameworks exist under various government departments and laws:

Major Institutions Involved in Technology Transfer

Institution	Role
Ministry of Science and Technology (MoST)	Policy direction and funding for R&D.
Bangladesh Council of Scientific and Industrial Research (BCSIR)	Research, innovation, and technology transfer to industries.
Bangladesh Investment Development Authority (BIDA)	Regulates foreign technology transfer in investment proposals.
Bangladesh University of Engineering and Technology (BUET)	Active in research and tech transfer projects.
ICT Division	Supports ICT-based innovations and startups.
Ministry of Industries (MoInd)	Promotes industrial development via TT initiatives.

Relevant Policies Supporting Technology Transfer

- 1. National Science and Technology Policy (NSTP) – 2011 (Updated Draft 2023)**
 - Encourages commercialization of research.
 - Calls for better IPR protection and licensing systems.
 - Proposes technology transfer units in universities and research centers.
- 2. National Industrial Policy – 2022**
 - Promotes technology-driven industry.
 - Supports joint ventures for technology acquisition.
- 3. Bangladesh Patent Law (Amended 2022)**
 - Strengthens IP protection, which is essential for safe tech transfer.
- 4. Export Policy & Import Policy Orders**
 - Include guidelines on the import of foreign technology.

Real-Life Example:

BCSIR's Spirulina Production Technology Transfer Background:

BCSIR developed indigenous technology for the production of **spirulina**, a high-protein nutritional supplement.

Transfer Process:

1. **Research & Development:** BCSIR developed cost-effective methods for spirulina cultivation.
2. **Licensing:** BCSIR licensed the technology to small entrepreneurs and agro-based industries.
3. **Training:** Farmers and producers were trained in production and drying techniques.
4. **Outcome:**
 - Empowered rural women entrepreneurs.
 - Reduced dependency on imported nutritional supplements.

Example of Foreign Technology Transfer to Bangladesh

Rooppur Nuclear Power Plant (RNPP)

Type: Foreign technology transfer via inter-governmental cooperation.

- **Russia's Rosatom** is transferring nuclear technology under strict regulatory oversight.
- Bangladesh Atomic Energy Regulatory Authority (BAERA) oversees safety and technology absorption.
- Includes technology training and local capacity building.

Challenges in NTTR Implementation:

- Bureaucratic delays in IP filings and licensing.
- Limited awareness among researchers.
- Inequitable access to technology for startups.
- Risk of unauthorized or insecure foreign transfers.

International Agreements (WTO-TRIPS, WIPO, Regional Policies):

International Agreements related to **Intellectual Property Rights (IPR)**, specifically focusing on:

- **WTO-TRIPS**
- **WIPO**
- **Regional Policies**

1. WTO-TRIPS Agreement

Overview:

- Administered by the **World Trade Organization (WTO)**.
- Came into effect in **1995** as part of the **Uruguay Round** negotiations.
- Sets **minimum standards** for protection and enforcement of various forms of IP (e.g., copyrights, patents, trademarks).

Key Features:

- Covers **patents, copyrights, trademarks, geographical indications, industrial designs, trade secrets, and layout designs of integrated circuits.**
- Applies to all WTO member countries.
- Encourages **technology transfer and innovation.**

Example: India's Patent Act Amendment (2005):

- India amended its Patent Act to become **TRIPS-compliant.**
- Before this, India allowed only **process patents** for pharmaceuticals, not product patents.
- After TRIPS, **product patents were introduced**, which meant drugs themselves could be patented.

2. WIPO (World Intellectual Property Organization)

Overview:

- A UN **specialized agency** founded in **1967.**
- Promotes the protection of intellectual property globally through cooperation among nations.

Functions:

- Administers **26 international treaties.**
- Provides a platform for **filing international patents** and trademarks (e.g., PCT and Madrid systems).
- Offers **IP-related services** like dispute resolution, technical assistance, and training.

Important Treaties Administered:

- **Paris Convention** (for the protection of industrial property)
- **Berne Convention** (for the protection of literary and artistic works)
- **Patent Cooperation Treaty (PCT)**
- **Madrid Agreement** (for international trademark registration) **Example: PCT (Patent Cooperation Treaty) System:**

◦ If an inventor wants patent protection in multiple countries, they can file a **single international application** through WIPO. ◦ E.g., A researcher in Germany can file under PCT and get patent protection in the US, Japan, and India with one application.

3. Regional Policies and Agreements

Overview:

- Several regions have their own IPR agreements to harmonize protection across member countries.

Examples:

a. European Union (EU) – European Patent Office (EPO):

- Provides a **centralized system** for obtaining patents across EU member states.
- A single application to the **EPO** can grant patents in multiple European countries.

b. African Regional Intellectual Property Organization (ARIPO):

- Promotes cooperation in IP matters among African countries.
- Supports **common patent and trademark procedures**.

c. Andean Community:

- Comprises **Bolivia, Colombia, Ecuador, and Peru**.
- Has **harmonized IP laws**, especially on biodiversity and traditional knowledge.

Case Studies on Successful Technology Transfer Policies:

1. Bayh-Dole Act (USA, 1980)

Policy Overview:

The **Bayh-Dole Act** allowed universities, small businesses, and non-profits in the U.S. to retain ownership of inventions made under federally funded research programs.

Success Mechanisms:

- Encouraged patenting of university research.
- Facilitated the licensing of university-developed technologies to private companies.
- Led to the creation of Technology Transfer Offices (TTOs) at universities. **Example:**
Stanford University & Google
- *Technology:* PageRank Algorithm developed by Larry Page and Sergey Brin while at Stanford.
- *Transfer Mechanism:* Licensed by Stanford's Office of Technology Licensing (OTL) to the founders.
- *Impact:* Led to the founding of **Google**, a trillion-dollar company.

2. Fraunhofer Institutes (Germany)

Policy Overview:

The **Fraunhofer Society** operates with a mission to apply research for industrial benefit, balancing public funding with industry contracts.

Success Mechanisms:

- Focus on applied research aligned with industry needs.
- Institutes own IP and commercialize it through licensing or startups.
- Revenues reinvested into R&D.

Example: Fraunhofer & MP3 Technology

- *Technology:* MP3 audio compression developed by Fraunhofer IIS.
- *Transfer Mechanism:* Licensed to companies like Microsoft, Apple, and others.
- *Impact:* MP3 became a global audio standard, generating over \$100 million in licensing revenues.

3. China's Torch Program (1988) Policy

Overview:

The **Torch Program** was initiated to promote high-tech industry development through tech transfer from research institutes.

Success Mechanisms:

- Created high-tech industrial development zones.
- Provided incentives for R&D commercialization.
- Encouraged collaboration between academia, research institutes, and enterprises.

Example: Huawei Technologies

- *Technology:* Built on collaborative research from Chinese universities.
- *Transfer Mechanism:* Support from Torch Program zones and university partnerships.
- *Impact:* Became a global telecom leader.

Role of Government in Innovation Policy

Governments play a pivotal role in driving innovation by funding research, building infrastructure, ensuring supportive policies, and enabling collaboration.

1. Funding Research & Development (R&D)

- **Description:** Grants, subsidies, and public research institutes support basic and applied research.
- **Example: USA – DARPA** funded internet (ARPANET), GPS, and AI.
- **Impact:** Provided foundations for private sector innovations (e.g., Google Maps).

2. Creating Innovation Infrastructure

- **Description:** Establishing science parks, incubators, R&D labs, and tech zones.
- **Example: India – STPI** offered infrastructure, tax incentives, and bandwidth.
- **Impact:** Enabled rise of India's software export industry (Infosys, Wipro).

3. Supportive Legal & Regulatory Frameworks

- **Description:** Protect IP rights, simplify business rules, set standards.
- **Example: EU – GDPR** created strict data privacy framework.
- **Impact:** Stimulated innovations in privacy-preserving technologies.

4. Tax Incentives & Subsidies

- **Description:** R&D credits, tax deductions, and subsidies reduce innovation costs.
- **Example:** Australia – **R&D Tax Incentive** (up to 43.5% refundable tax offset).
- **Impact:** Boosted R&D investment among startups and SMEs.

5. Public-Private Partnerships (PPP)

- **Description:** Government co-invests with industry to share innovation risk.
- **Example:** Germany – **Industrie 4.0** initiative for smart manufacturing.
- **Impact:** Advanced automation, IoT, and smart factories.

6. Education & Skill Development

- **Description:** Funding STEM education, vocational training, and research fellowships.
- **Example:** South Korea – **STEAM Education Policy** (Science, Tech, Engineering, Arts, Math).
- **Impact:** Strengthened innovation economy (Samsung, Hyundai).

7. Technology Transfer & Commercialization

- **Description:** Support through tech transfer offices, licensing, and startup incubation.
- **Example:** Japan – **METI** helps startups commercialize university research.
- **Impact:** Growth of deep-tech startups and spinoffs.

8. International Collaboration & Innovation Diplomacy

- **Description:** Cross-border R&D programs and global partnerships.
- **Example:** EU – **Horizon Europe Program**, the world's largest multinational research fund.
- **Impact:** Supports global R&D in AI, energy, health, etc.